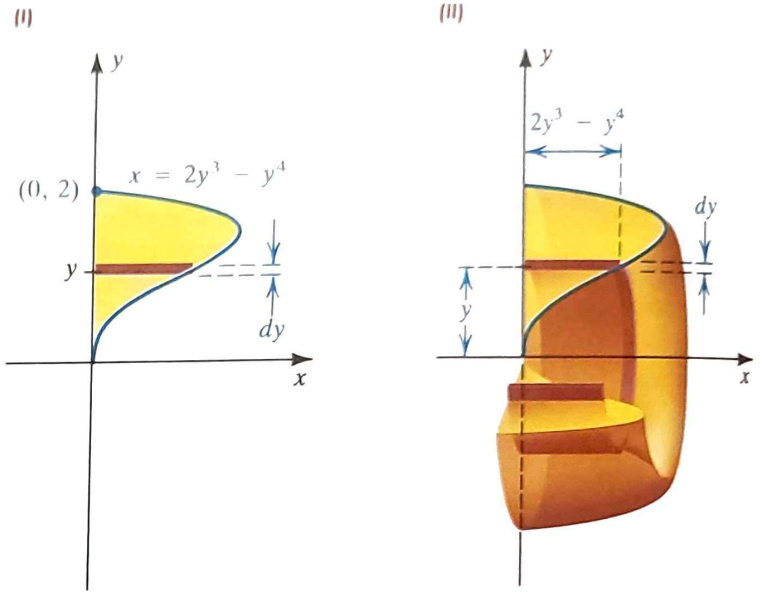
FIGURE 6.33



To sum all such shells, we move upward through the region from $c =$ to $d = 2$. Hence the limit of sums is

$$V = \int_0^2 2\pi y(2y^3 - y^4)\, dy.$$

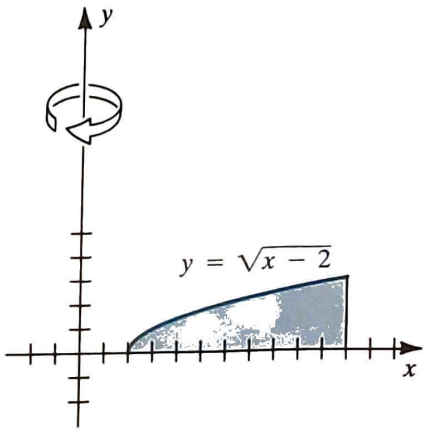
It is worth noting that in the preceding example we were forced to use shells and to integrate with respect to y, since the use of washers and integration with respect to x would require that we solve the equation $x = 2y^3 - y^4$ for y in terms of x, a rather formidable task.

EXERCISES 6.3

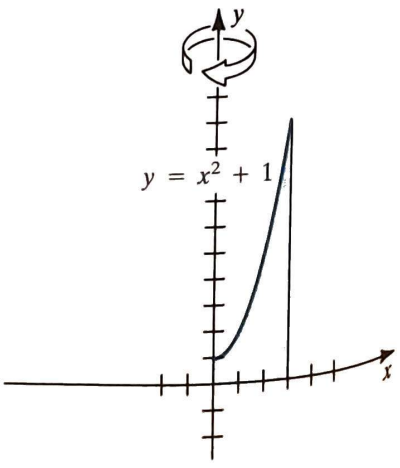
Use cylindrical shells for each exercise.

Exer. 1–4: Set up an integral that can be used to find the volume of the solid obtained by revolving the shaded region about the indicated axis.

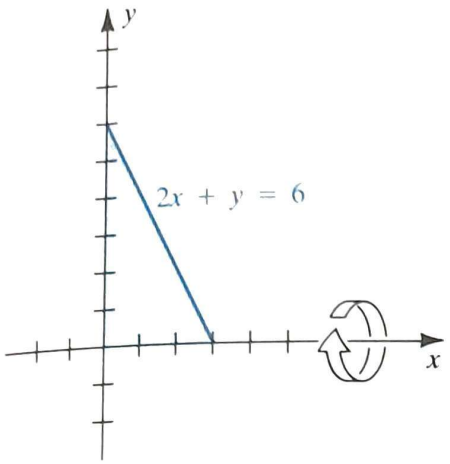
1



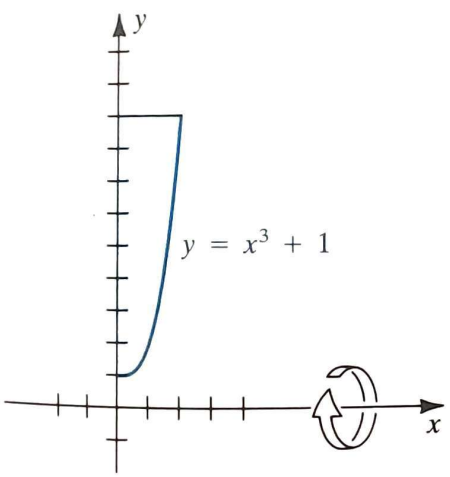
2



3



4



Exer. 5–18: Sketch the region R bounded by the graphs of the equations, and find the volume of the solid generated if R is revolved about the indicated axis.

5 $y = \sqrt{x}$, $x = 4$, $y = 0$; y-axis

6 $y = 1/x$, $x = 1$, $x = 2$, $y = 0$; y-axis

7 $y = x^2$, $y^2 = 8x$; y-axis

8 $16y = x^2$, $y^2 = 2x$; y-axis

9 $2x - y = 12$, $x - 2y = 3$, $x = 4$; y-axis

10 $y = x^3 + 1$, $x + 2y = 2$, $x = 1$; y-axis

11 $2x - y = 4$, $x = 0$, $y = 0$; y-axis

12 $y = x^2 - 5x$, $y = 0$; y-axis

13 $x^2 = 4y$, $y = 4$; x-axis

14 $y^3 = x$, $y = 3$, $x = 0$; x-axis

15 $y = 2x$, $y = 6$, $x = 0$; x-axis

16 $2y = x$, $y = 4$, $x = 1$; x-axis

17 $y = \sqrt{x + 4}$, $y = 0$, $x = 0$; x-axis

18 $y = -x$, $x - y = -4$, $y = 0$; x-axis

Exer. 19–26: Let R be the region bounded by the graphs of the equations. Set up an integral that can be used to find the volume of the solid generated if R is revolved about the given line.

19 $y = x^2 + 1$, $x = 0$, $x = 2$, $y = 0$
(a) $x = 3$ (b) $x = -1$

20 $y = 4 - x^2$, $y = 0$
(a) $x = 2$ (b) $x = -3$

21 $y = x^2$, $y = 4$
(a) $y = 4$ (b) $y = 5$ (c) $x = 2$ (d) $x = -3$

22 $y = \sqrt{x}$, $y = 0$, $x = 4$
(a) $x = 4$ (b) $x = 6$ (c) $y = 2$ (d) $y = -4$

23 $x + y = 3$, $y + x^2 = 3$; $x = 2$

24 $y = 1 - x^2$, $x - y = 1$; $y = 3$

25 $x^2 + y^2 = 1$; $x = 5$

26 $y = x^{2/3}$, $y = x^2$; $y = -1$

Exer. 27–30: Let R be the region bounded by the graphs of the equations. Set up integrals that can be used to find the volume of the solid generated if R is revolved about the given axis using (a) cylindrical shells and (b) disks or washers.

27 $y = 1/\sqrt{x}$, $x = 1$, $x = 4$, $y = 0$; x-axis

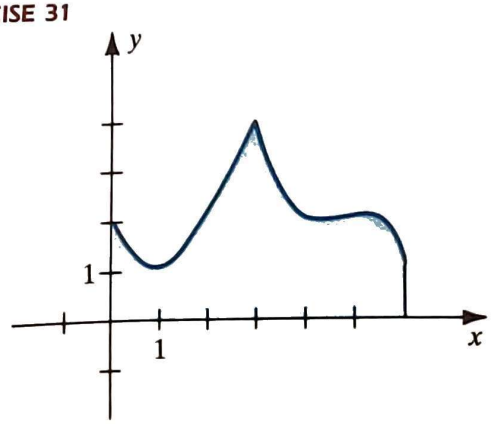
28 $y = 9 - x^2$, $x = 0$, $x = 2$, $y = 0$; x-axis

29 $y = x^2 + 2$, $x = 0$, $x = 1$, $y = 0$; y-axis

30 $y = x + 1$, $x = 0$, $x = 1$, $y = 0$; y-axis

31 If the region shown in the figure is revolved about the y-axis, use the trapezoidal rule, with $n = 6$, to approximate the volume of the resulting solid.

EXERCISE 31



32 If the region shown in the figure on the following page is revolved about the y-axis, use Simpson's rule, with $n = 8$, to approximate the volume of the resulting solid.